Power BI Row-Level Security (RLS) Guide

Data Modeling Concepts, Relationship Mechanics & Implementation Steps

1. Diagnostic & Core Data Concepts

The One-to-Many Relationship Rule

In tabular data modeling (Power BI / Analysis Services), creating a **One-to-Many (1:*)** relationship requires that the column on the "1" side contains **100% unique values** with zero duplicates.

WHY STANDARD RELATIONSHIP CREATION FAILED

Attempting to connect `Users[Region]` to `Product[Region]` fails as a 1:* relationship because your `Users` table contains duplicate region entries (e.g., multiple users located in UAE, `India`, or UK). Power BI cannot determine which single row to reference.

2. Data Model Architectural Solutions

Approach A: Static RLS via Direct Table Rules (Easiest / Starter Method)

STATIC SECURITY Best for simple role assignments where users belong to fixed, non-changing regional groups.

Instead of linking a security table in the data model, rules are defined directly on the data table (e.g., `Product`) using DAX filters.

```
+--------------------+
|      Product       |  (Filtered directly via DAX Role definition)
+--------------------+
| ProductID          |
| ProductName        |
| Category           |
| Region   <------------ DAX Filter: [Region] = "UAE"
| Sales              |
+--------------------+
```

Approach B: Normalized Dimension Table (Star Schema Method)

To avoid direct many-to-many relationship issues, extract unique regions into a standalone master dimension table.

DAX to create unique table:

```
Region = DISTINCT(Product[Region])
```

```
  Users                    Region                 Product
+----------+             +--------+             +-------------+
| UserID   |             | Region | (Unique 1)  | ProductID   |
```

```
| UserName |             +--------+             | ProductName |
| Email    |                 |                  | Category    |
| Region   | (*)             | (1)              | Region  (*) |
+----------+                 |                  | Sales       |
     |                       |                  +-------------+
     +---- (Many) ----------->                         ^
                                                       |
                                  (1:Many Relationship) --+
```

Approach C: Enterprise Dynamic Security Model (Bridge Pattern)

DYNAMIC SECURITY Scalable pattern for complex enterprise scenarios where single users access multiple regions.

```
  [Users]            [UserRegion]            [Region]           [Product]
+----------+        +--------------+        +--------+        +-------------+
| UserID   | (1:* ) | UserEmail    |  ( *:* )| Region | (1:* ) | ProductID   |
| Email    |------->| Region       |-------->|        |------->| ProductName |
+----------+        +--------------+        +--------+        | Region      |
                                                              +-------------+
```

3. Step-by-Step Implementation Instructions

Option 1: Setting Up Direct Static Roles (Recommended for Learning)

1. Open **Power BI Desktop** with your dataset loaded.
2. Navigate to the **Modeling** tab in the top ribbon bar.
3. Click on **Manage Roles**.
4. In the Roles window, click **Create** and enter the name: `UAE_Role`.
5. Under **Tables**, select the `Product` table.
6. In the Table Filter DAX expression box, enter:

```
[Region] = "UAE"
```

7. Click **Save**.
8. Repeat the steps to create:
   - `India_Role` with filter: `[Region] = "India"`
   - `UK_Role` with filter: `[Region] = "UK"`
9. **Verification:** Go to **Modeling** → **View as**, check `UAE_Role`, and click **OK**. The visual tables will dynamically filter down to display only UAE region data.

Option 2: Setting Up Dynamic RLS using User Principal Name

When deploying to the Power BI Service for automated user-level filtering, use system functions:

1. In **Manage Roles**, create a role named `Dynamic_User_Role`.
2. Select the `Users` table and apply the following DAX expression:

```
[Email] = USERPRINCIPALNAME()
```

3. Ensure bi-directional security filtering is enabled along the relationship path to allow the `Users` table filter to flow to the `Product` table.

4. Summary Comparison Matrix

Method	Complexity	Maintenance Effort	Best Used For
Direct Static Roles	Low	High (Manual role updates per user)	Simple reports, static regional assignments, learning basics.
Distinct Region Table	Medium	Medium	Clean Star-Schema modeling; standard regional reporting.
Dynamic UPN Model	Advanced	Low (Automated via Active Directory / Azure AD)	Enterprise deployments, large user bases, complex security rules.